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| **PI Name:** | Blakely, Eleanor A Ph.D. | | |
| **Project Title:** | Early Markers of Space Radiation Induced Cataractogenesis | | |
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| | Cataract incidence in astronauts has been correlated with exposure to relatively low absorbed doses of radiation in space flight. The type of space radiation linked with low-dose cataractogenesis has not yet been identified. The goal of this proposal is to test the hypothesis that radiation-induced alterations in early molecular signaling in cultured human lens epithelial cells (HLE) can be used to identify early markers of relative risk of cataracts from key high-LET radiation components. We have shown that exposure to low fluences of high-LET radiation promotes severe lens cell dysfunction via events that include modulation of FGF-2, inhibition of cell cycle control via cyclin-dependent-kinase inhibitors (CDKIs) altered expression of cell receptor and adhesion molecules, and altered apoptotic/enucleation program machinery, in the HLE and human lens fiber (HLF) cells. These observations are consistent with a model in which ionizing radiation initiates premature differentiation. We propose three specific aims that will use a matrix of charged particles of high energy and atomic number to test important aspects of our hypothesis, the link to cataractogenesis, and that will also allow the development of biological countermeasures. Quantitative measurements of the inappropriate | | |

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| **Task Description:** | commencement of fiber cell differentiation will be made as a function of particle fluence, atomic number and energy, and correlated to the kinetics of early signaling expression in HLE cells in vitro, and in the nascent radiation-induced cataract in the rat in vivo. The human lens is one of the most radiosensitive organs of the body, and is one of the tissues upon which shielding requirements for space flight have been based to avoid cataracts, a late-appearing radiation effect. Recent evidence however indicates that exposure to relatively low doses of space radiation is associated with an increased incidence and earlier appearance of human cataracts than was anticipated. We developed an in vitro model of differentiating human lens epithelial (HLE) cells, and have been studying their response to different types of ionizing radiations prevalent in space travel under conditions that mimicked the normal growth environment in the tissue, but were more amenable to investigation than in vivo models. The objective to our project is to provide evidence showing that particle radiation induces alterations in gene expression profiles when compared to the normal lens differentiation process resulting in premature differentiation subsequently leading to cataract formation. Historically, radiation-induced cataract has been anatomically characterized as posterior sub-capsular (PSC) due to the initial location of its appearance within the lens after radiation exposure. The mechanisms underlying radiation-induced cataract are thought to involve direct DNA damage to the epithelial cells, which is later expressed as an opacification when the epithelial cells have differentiated into fiber cells and migrated to the PSC location. However lens fiber cells at the juncture of the posterior capsule and suture are terminally differentiated in the region where fibers cells are devoid of nuclear organelles including the cell nucleus. The radiation-induced cataract is a progressive lesion that continues to evolve during the life of the organism, and the molecular basis for the origin and development of radiation-induced cataract are largely unknown. The continuous lifetime migration of germinative epithelial cells down the surface of the anterior surface of the lens of the eye, and the progressive transition into primary and secondary fiber cells involves a coordinated symphony of changes. The results from our work will provide information on the important roles of several of radiation-responsive signal transduction pathways in lens differentiation and of pre-cataractous changes. Understanding of the molecular basis in the cataract pathology will provide important tools to develop biological approaches for countermeasures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our cell culture model is an exciting tool that demonstrates human lens epithelial cell growth on an extracellular matrix derived from bovine corneal endothelial cells, and also exhibits aspects of fiber cell differentiation with the expression of several morphological and protein markers. The results of our work will be used to pursue countermeasures that could reduce the risk of cataract. These studies may also contribute to estimations of risk assessment for radiation-induced cataracts among astronauts exposed to particles during space travel, especially during missions into deep space. We may also contribute to the development of countermeasures for radiation-induced cataracts from other occupational and clinical radiation exposures. The biological mechanisms underlying normal lens fiber cell differentiation may also be elucidated. The exact factor(s) responsible for the continued replication of lens epithelial cells in embryos and adults remains to be determined. These studies therefore should shed new light on the process of human lens cell differentiation, that has been difficult to explore with animal models in vivo or with human lens cultures in vitro that are either immortalized and/of undergo only limited differentiation. |
| **Task Progress:** | A focus of our work during this year was to characterize the expression of the cycle-dependent-kinase inhibitor CDKN1A, also known as p21WAF1/CIP1. This protein is known to play a critical role in the regulation of cell cycle kinetics, DNA synthesis, and stress responses leading to apoptotic cell death. This protein has also been demonstrated to play a role in several pathways functionally related to LET-responsive radiation damage. A lesser-known function of p21 has been linked to stem cell differentiation in a number of cell lineages. Through our collaboration with Drs. L. Goldstein, L. Chylack and R. Mancini at the Brigham and Womens Hospital, MA, we demonstrated the nascent localization of p21 protein in the transitional zone around the bow region of normal human lens tissues suggesting that p21 plays a functional role in the differentiation of lens epithelial cells into fiber cells.<br>Differentiating human lens epithelial cells (HLE) were exposed to 1 GeV/amu Iron ions (LET=148 keV/micrometer) at the NASA Space Radiation Laboratory (NSRL), Protons (LET = 1.2 keV/micrometer) at the 88 inch cyclotron or 150 kVp Xrays at the Lawrence Berkeley National Laboratory (LBNL).We quantitatively assessed the transcriptional and translational regulation of p21 gene products in a time course before and after single 4 Gy radiation doses.<br><br>Immunofluorescence studies show that although all the radiations studies induced nuclear-localized upregulated p21 expression, the magnitude of enhancement appears to be LET-dependent. Quantitative analyses of signal intensities in p21-positive cells show that 4 Gy of Xrays induced an overall increase in p21-positive signals in the cells, whereas equidose of proton irradiation, resulted in a shift of the spectrum of p21 positive cells to a higher level. Iron ions induced an intense overdispersion of fluorescent signals in HLE cells suggesting a broader range of enhanced p21 protein expression. Western analysis of samples obtained from parallel cultures indicated p21 expression was increased approximately 1.3 fold after iron ion exposure when compared to unirradiated controls. X-ray and proton results were similar and not significantly different from those obtained from the controls. This diminished overall magnitude of the increased p21 protein expression measured may be due to population averaging of the wide distribution of variable p21-expressing cells.<br><br>Using the focused gene array approach to examine radiation-induced alterations in transcriptional regulation of genes known to be involved in the cell cycle control, RNA from Control and 4 Gy proton irradiated samples were used as probes to detect the expression of 96 genes known to be involved in cell cycle kinetics. Our results show definitive differences between the control and irradiated samples, with p21 among one of the upregulated genes. We extended our studies further by quantitatively evaluating p21 transciption as a function of time after a single 4 Gy dose of Xrays, protons or Iron ions. We observed a transient upregulation of p21 transcripts between 2 to 4h after exposure for all three beams. The magnitude of induction was LET-dependent. p21 transcription was transiently elevated up to 4-fold above controls at 3 hr after x-ray exposure, and elevated up to >5 fold above control level at 3 to 4 hr after proton irradiation. Iron ions induced a much higher level of p21 expression as early as 2 h after exposure. Although we saw evidence of a downturn in transcriptional activity at about 4h after exposure, p21 expression at 6 and 8h after exposure show up 10 to 100 fold enhanced expression. These novel findings suggest that the impact of iron ions irradiation on p21 may be very different from those obtained from low LET particles. With our differentiating lens culture model, we have evidence showing that the differentiation of lens epithelial cells into fiber cells is altered by exposure to ionizing radiations. Fiber cells elongate, and begin alignment with each other prematurely when epithelial cells are irradiated, and fiber cell specific adhesion markers are expressed prematurely (Blakely et al., in preparation). Based on our evidence, our hypothesis is that p21 gene and protein expression has an important role in several of these radiation-responsive signal |

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| | transduction pathways linked to differentiation and may provide molecular evidence of pre-cataractous changes, as well as a biological approach to develop countermeasures.<br><br>During the NSRL-2 campaign in April 2004, we initiated a pilot study aimed to compare our in vitro finding to an in vivo model, through immunohistochemical evaluations. Sprague Dawley rats were exposed to a low (10 cGy) or a high (100 cGy) dose of 600 MV/nucleon Iron ions or 1 GeV/nucleon Titanium ions. Cataract formation in these animals is being monitored on a regular basis using slip lamp systems developed by our collaborators, Drs. Chylack, Clark, Seeberger and Gratzek. Results from these studies will be reported in future taskbooks. |
| **Bibliography Type:** | Description: (Last Updated: 02/12/2026) |
| **Abstracts for Journals and Proceedings** | Blakely EA, Chang PY, Bjornstad KA, McNamara MP, Rosen CJ, Chylack LT, Goldstein LE. "Profile of ß-amyloid precursor protein (ß-APP) synthesis and processing during human lens cell differentiation." Annual Meeting of Association of Research in Vision and Ophthalmology (ARVO), Fort Lauderdale, FL, April 2004.<br>Annual Meeting of Association of Research in Vision and Ophthalmology (ARVO), Fort Lauderdale, FL, April 2004. Abstract # 2627. , Apr-2004 |
| **Presentation** | Blakely, E.A. "Space Radiation and Cataract." Lawrence Berkeley Laboraotory Summer Colloquium, Berkeley, CA.<br>Jun-2003 |
| **Presentation** | Blakely, E.A. "Molecular Biology of Radiation-induced Cataractogenesis." COSPAR Colloquium (Second International Workshop) Radiation Safety for Manned Mission to Mars, Dubna, Russia.<br>Sep-2003 |
| **Presentation** | Blakely, E.A. "Space Radiation and Cataract." Northern California Chapter of Health Physics Society, at Genentech, South San Francisco, CA.<br>Nov-2003 |
| **Presentation** | Blakely, E.A.; Chang, P.Y.; Bjornstad, K.A.; Rosen, C.J.; McNamara, M.P., Mancini, R.; Goldstein, L.E.; Chylack, L.T. "Iron ion-, Proton- and Xray Effects on Human Lens Cell Differentiation." 3rd International Workshop on Space Radiation Research being held in conjunction with the 15th Annual NASA Space Radiation Health Investigators' Workshop.<br>May-2004 |
| **Presentation** | Blakely, E.A.; Chang, P.Y.; Bjornstad, K.A.; Rosen, C.J.; McNamara, M.P.; Mancini, R.; Goldstein, L.E.; Chylack, L.T. "Iron ion-, proton- and x-ray-effects on human lens." Radiation Research Annual Meeting, St Louis, MO.<br>Apr-2004 |